# Practice workbook · Finding stationary points

Setting dy/dx = 0 and solving (quadratics, factorised cubics or trig equations) to find the coordinates of stationary points.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-CAL-08

# 1. Question 7 · part a · 5 marks

2021 JUN · Paper 01

Worked-example retry · Attempt part a only.

**7** The curve $C$ has equation $y = \dfrac{x}{x^2 + 4}$

(a) Using calculus, find the coordinates of the stationary points on $C$.

**(5)**

(b) Show that $\dfrac{d^2y}{dx^2} = \dfrac{2x(x^2 - 12)}{(x^2 + 4)^3}$

**(4)**

(c) Hence, or otherwise, determine the nature of each of these stationary points.

**(2)**

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## 1. Question 7 · part a · 5 marks · Working and answer

Working and answer

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## 2. Question 9 · part b.ii · 7 marks

2019 JUN · Paper 02

Further practice · Attempt part b.ii only.

**9** The curve $C$, with equation $y = f(x)$, passes through the point with coordinates $\left(-2, -\frac{28}{3}\right)$

Given that $f'(x) = x^3 - x^2 - 4x + 4$

(a) show that $C$ passes through the origin.

**(4)**

(b) (i) Show that $C$ has a minimum point at $x = 2$ and a maximum point at $x = 1$

(ii) Find the exact value of the $y$ coordinate at each of these points.

**(7)**

The curve has another turning point at $A$.

(c) (i) Find the coordinates of $A$.

(ii) Determine the nature of this turning point.

**(3)**

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## 2. Question 9 · part b.ii · 7 marks · Working and answer

Working and answer

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## 3. Question 9 · part c.i · 0 marks

2019 JUN · Paper 02

Further practice · Attempt part c.i only.

**9** The curve $C$, with equation $y = f(x)$, passes through the point with coordinates $\left(-2, -\frac{28}{3}\right)$

Given that $f'(x) = x^3 - x^2 - 4x + 4$

(a) show that $C$ passes through the origin.

**(4)**

(b) (i) Show that $C$ has a minimum point at $x = 2$ and a maximum point at $x = 1$

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The curve has another turning point at $A$.

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## 3. Question 9 · part c.i · 0 marks · Working and answer

Working and answer

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## 4. Question 9 · part c · 6 marks

2020 NOV · Paper 02

Further practice · Attempt part c only.

**9** A curve $C$ has equation

$$y = \frac{2 + 4x - x^2}{2x + 1} \quad x \neq -\frac{1}{2}$$

(a) Write the equation of $C$ in the form $ax^2 + (by - 4)x + (y - c) = 0$, where $a$, $b$ and $c$ are integers whose values are to be found.

**(3)**

(b) Hence show that $x$ is real when $y \leqslant 2$ and when $y \geqslant 3$

**(4)**

(c) Find the coordinates of the stationary points on $C$.

**(6)**

(d) Sketch $C$ showing clearly

(i) the exact coordinates of the points where $C$ crosses the $x$-axis,

(ii) the asymptote to $C$ that is parallel to the $y$-axis,

(iii) the coordinates of the stationary points.

**(5)**

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P 6 2 2 8 5 A 0 2 4 3 2

## 4. Question 9 · part c · 6 marks · Working and answer

Working and answer

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## 5. Question 11 · part c · 7 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

**11** (a) Using a formula on page 2 show that $\cos 2A = 2\cos^2 A - 1$

(2)

(b) Hence show that $\left(2\cos^2 A - 1\right)^2 = \dfrac{\cos 4A + 1}{2}$

(3)

The curve with equation $y = \dfrac{\sin 2x}{2} + \dfrac{\left(2\cos^2 x - 1\right)^2}{2} + \dfrac{1}{8}$ has a stationary point $P$ in the range $0 \leqslant x \leqslant \dfrac{\pi}{6}$

(c) Find the exact coordinates of $P$

(7)

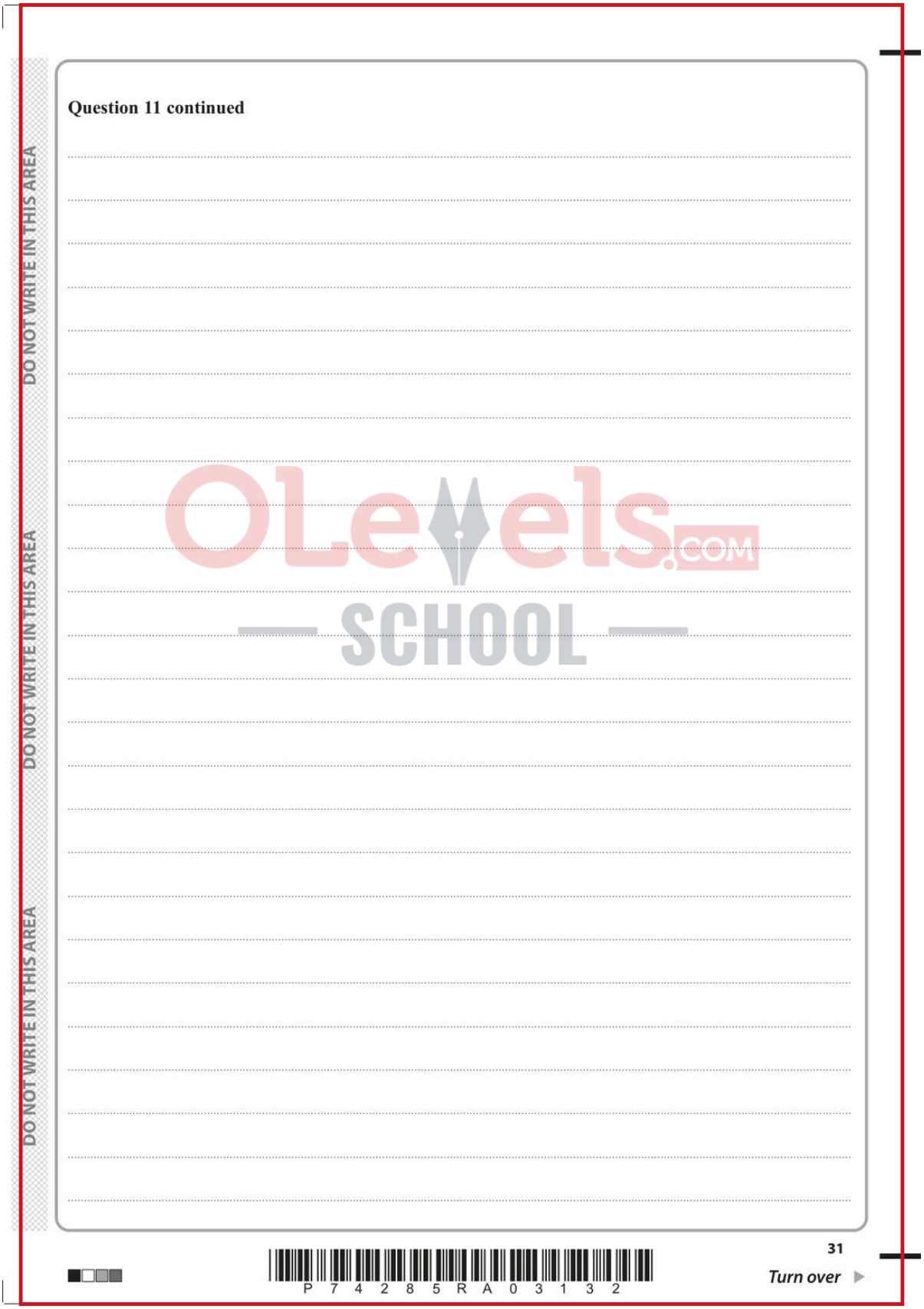
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## 5. Question 11 · part c · 7 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.



## 5. Question 11 · part c · 7 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

**Question 11 continued**

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**(Total for Question 11 is 12 marks)**

## 5. Question 11 · part c · 7 marks · Working and answer

Working and answer

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## 6. Question 7 · part b.i · 5 marks

2022 JUN · Paper 01

Further practice · Attempt part b.i only.

**7** The point with coordinates $(4, -104)$ lies on the curve $C$ with equation $y = f(x)$

Given that $f'(x) = 4x^3 - 12x^2 - 19x + 12$

(a) (i) show that $C$ passes through the origin,

**(4)**

(ii) show that $C$ has a maximum at the point on the curve where $x = 0.5$

**(3)**

The curve $C$ has another turning point at $A$ and another turning point at $B$.

Given that the $x$ coordinate of $A$ is negative,

(b) (i) find the coordinates of $A$ and the coordinates of $B$,

**(5)**

(ii) determine the nature of these turning points.

**(3)**

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**20**

P 7 1 6 6 1 A 0 2 0 3 2

## 6. Question 7 · part b.i · 5 marks · Working and answer

Working and answer

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## 7. Question 10 · part b · 4 marks

2021 NOV · Paper 01

Further practice · Attempt part b only.

**10**

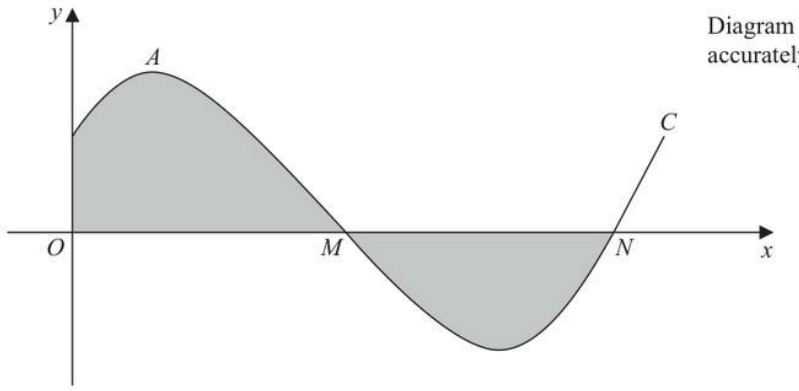



Diagram **NOT** accurately drawn

**Figure 4**

Figure 4 shows the curve $C$ with equation $y = \frac{1}{2} + \sin 3x$ where $0 \leqslant x < \frac{2\pi}{3}$

The curve $C$ crosses the $x$-axis at the points $M$ and $N$

(a) Show that the coordinates of $M$ are $\left(\frac{7\pi}{18}, 0\right)$ and find the coordinates of $N$

**(3)**

The curve $C$ has a maximum at the point $A$

(b) Find the coordinates of $A$

**(4)**

(c) Find an equation of the tangent to $C$ at $M$

Give your answer in the form $ay + b\sqrt{3}x - c\sqrt{3}\pi = 0$ where $a$, $b$ and $c$ are integers to be found.

**(4)**

The finite region, shown shaded in Figure 4, is bounded by the curve $C$, the $y$-axis and the part of the $x$-axis from $O$ to $N$

(d) Use algebraic integration to find, to 3 significant figures, the total area of the shaded region.

**(4)**

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## 7. Question 10 · part b · 4 marks · Working and answer

Working and answer

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## 8. Question 4 · part a · 4 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part a only.

**4** The curve $C$ has equation

$$y = x^3 - 3x^2 - 24x + 6$$

(a) Use calculus to find the coordinates of each of the stationary points on $C$.

**(4)**

(b) Determine the nature of each of these stationary points.
Justify your answers.

**(2)**

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## 8. Question 4 · part a · 4 marks · Working and answer

Working and answer

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